

KETHNI NANAYAKKARA

Software Engineering | AI/ML Contributor to WSO2 Open Source

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PROFESSIONAL SUMMARY

I care about AI that adapts to people, not the other way around — building on-device, privacy-first, and culturally aware systems that stay useful in everyday, real-world contexts. As a Software Engineering undergraduate at NSBM Green University (University of Plymouth, UK), I apply object-oriented programming and back-end development principles through merged contributions to WSO2 production repositories, API-driven full-stack projects, and EmoUI, a real-time AI-powered Chrome extension published on the Chrome Web Store. Experienced in debugging, code review, and remote/agile team collaboration across academic, research, and open-source projects.

EDUCATION

BSc (Hons) Software Engineering 2023 – Expected Graduation: Dec 2026

NSBM Green University – University of Plymouth, UK

Foundation Programme in Computing 2023 – Oct 2023

NSBM Green University

TECHNICAL SKILLS

AI / ML: CNN, YOLO (Ultralytics), MediaPipe, TensorFlow Lite, OpenCV, scikit-learn, Generative AI Tools

Languages: JavaScript, Python, SQL

Frontend: React, HTML5, CSS3, Tailwind CSS

Backend: Node.js, Express.js

Databases: MongoDB, MySQL, Firestore

Tools & Practices: Git, GitHub, Agile/Scrum, RESTful APIs, Object-Oriented Programming (OOP), Debugging & Code Review

Debugging & Code Review

KEY PROJECTS

EmoUI – Emotion-Adaptive Chrome Extension 2025 – 2026

Individual Final Year Project | [Live on Chrome Web Store](#)

Tech: Python, React, Tailwind CSS, OpenCV, YOLO11n-cls, MediaPipe

- Built a real-time, privacy-first, on-device facial emotion recognition Chrome extension with adaptive UI themes, audio feedback, and motivational quotes in English and Sinhala.
- Benchmarked 3 ML architectures against each other on a multi-source dataset, selecting and deploying YOLO11n-cls (94.2% Top-1 accuracy) into a production Chrome extension used by real users.
- Published to the Chrome Web Store; associated comparative research paper submitted to ICTAR 2026.

Hotel Booking & Reservation System 2026

Full-Stack Group Project

Tech: React, Node.js, Express.js, MongoDB

- Built a full-stack platform for hotel listings, room availability, and reservation management using collaborative Git workflows and Agile practices.

FurniViz – Interactive Furniture & Room Visualizer 2026

Group Project | *Role: UX Lead*

Tech: React.js, Vite, Three.js, Fabric.js, Tailwind CSS

- Built an interactive web app with a real-time synchronized 2D grid planner and 3D viewer for room layout design.
- Designed interaction flows, applied Nielsen's usability heuristics, and conducted think-aloud usability testing.

Caring for Carers – Autism Caregiver App 2025

Mobile Group Project + Research

Tech: Flutter, Firebase, TensorFlow Lite, Gemini API

- Developed stateful UI components and a visual communication library with Firebase real-time integration.
- Contributed to an AI-assisted caregiver support system and co-authored a published conference abstract at KDU IRC 2025.

GlowMate – Skincare Recommendation App 2025

Tech: Flutter, Firebase Authentication, Firestore

- Implemented user authentication and Firestore-backed quiz result handling.

OPEN SOURCE CONTRIBUTIONS

Open Source Contributor | WSO2 (GitHub: Kethniilmasha) | 2025 – Present

- **identity-apps #10345 (Merged)** — Debugged and refactored nested conditional logic across 7 files in a production API resource management feature, collaborating remotely with a senior WSO2 reviewer through a formal code review process to merge the fix.
- **docs-is #6174 (Merged)** — Improved developer onboarding documentation for Asgardeo's identity platform by simplifying workflow steps, contributing to a distributed open-source team under an agile, remote review cycle.
- **docs-is #5916 (Merged)** — Fixed grammar and standardized technical casing in SMS/Email OTP docs and the Identity Server Quick Start Guide.
- **docs-is #5828(Merged)** — Corrected spelling errors and broken curl URL path placeholders in the Asgardeo Identity Governance API spec.
- **docs-apim #11470 (Merged)** — Fixed MkDocs list rendering in the API Manager Quick Start Guide.

CERTIFICATIONS

- **AI/ML Engineer – Stage 1** — SLIIT, Centre for Open and Distance Education (2026)
- **What Is Generative AI?** — Microsoft LinkedIn Learning (2026)
- **Front-End Web Development** — Centre for Open & Distance Learning (CODL), University of Moratuwa (2026)
- **Web Design for Beginners** — Centre for Open & Distance Learning (CODL), University of Moratuwa (2026)
- **React (Basic)** — HackerRank Skill Certification (2026)
- **Python (Basic)** — HackerRank Skill Certification (2026)

RESEARCH & PUBLICATIONS

- **ICTAR 2026 (Submitted Conference Paper):** Culturally Diverse, Demographically Aware Facial Emotion Recognition Using YOLO11n-Classification: A Comparative Study — associated with the EmoUI final-year project.
- **KDU IRC 2025 (Published Conference Abstract):** AI-assisted caregiver support system —co-author; contributed to system design and evaluation.

REFERENCES

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